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**COURSE: MACHINE LEARNING AND NEURAL NETWORKS**

# ACTIVATION FUNCTIONS IN NEURAL NETWORKS FOR HEART DISEASE PREDICTION

## 1.0 INTRODUCTION

Machine learning is a type of artificial intelligence (AI) approach widely used in healthcare for disease prediction and clinical decision support. Through the analysis of clinical data, the model can extract hidden patterns that may not be recognized by the clinician. Hence, enabling the doctor for early detection of diseases, effective treatment, and better healthcare for the patients.

Neural networks are one of the more advanced machine learning methods that can be employed in healthcare analytics. They are based on the structure and function of the human brain and have the capability to model highly complex non-linear relationships in the data. Examples of using neural networks in healthcare include disease diagnosis and prognosis, medical imaging, and patient risk assessment.

In this tutorial, we are going to talk about the activation functions and how they affect the neural network performance. An activation function is non-linear, which means that it allows the neural network to learn non-linear relationships. These relationships are not possible with a linear model.

In this tutorial, we will look at how different activation functions affect the performance of a neural network. We will be training a neural network model on a heart failure prediction dataset from Kaggle. The performance of the Sigmoid, Tanh and ReLU activation functions will be compared.

## 2.0 NEURAL NETWORKS

Neural networks are composed of many interconnected computing units called neurons. Each computing unit is akin to a simple computer and the simplest form of the neuron is known as a perceptron. A perceptron computes a weighted sum of the input values and applies an activation function to generate an output.

$$y = f(w_1x_1 + w_2x_2 + \ldots + b)$$

Neural networks are structured into layers:

**Input Layer**: Receives features such as age, cholesterol, and blood pressure.
**Hidden Layers**: Learn intermediate representations of the data.
**Output Layer:** Produces the final prediction.

When multiple hidden layers are used, the model is referred to as a deep neural network. Deep learning allows the models to learn at a deeper level in the data and as a result are often very powerful for complex classification and regression tasks.

## 2.1 ACTIVATION FUNCTIONS

Activation functions decide whether a neuron should be activated. If there wasn't any activation function in the neural network, then the neural network would end up acting as a linear model, and linear models can't represent the complex relationships in the data we're interested in.

Activation functions are the decision making gates of a network. The ReLU (Rectified Linear Unit) is a function which acts like a switch - it only allows current to pass through when the input is positive.

**Sigmoid**

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

The sigmoid function maps real values to the range (0, 1). It is popular in neural networks where the output of a node should be a probability. One of the drawbacks of the sigmoid function is that it leads to the vanishing gradient problem where gradients can become very small.

**Tanh**

$$\tanh(x) = \frac{e^{x} - e^{-x}}{e^{x} + e^{-x}}$$

The Tanh output ranges from -1 to 1 and is zero centred. So is potentially easier to optimise than sigmoid.

**ReLU**

$$ReLU(x) = \max(0, x)$$

ReLU stands for Rectified Linear Unit, a version of activation function, which not only helps in speeding up the computations but also helps in removing the vanishing gradient problem, that's why it is the most popular activation function in deep learning that is being used in the most of the models of the modern neural networks.

## 3.0 DATASET OVERVIEW

The heart failure prediction dataset has many clinical parameters about heart such as age, cholesterol level, resting blood pressure, max heart rate etc. This target parameter shows that a heart disease is present or not.

Most of the Datasets I used come from Kaggle. Kaggle is actually a great place to do some machine learning tutorials and benchmark your experiments.

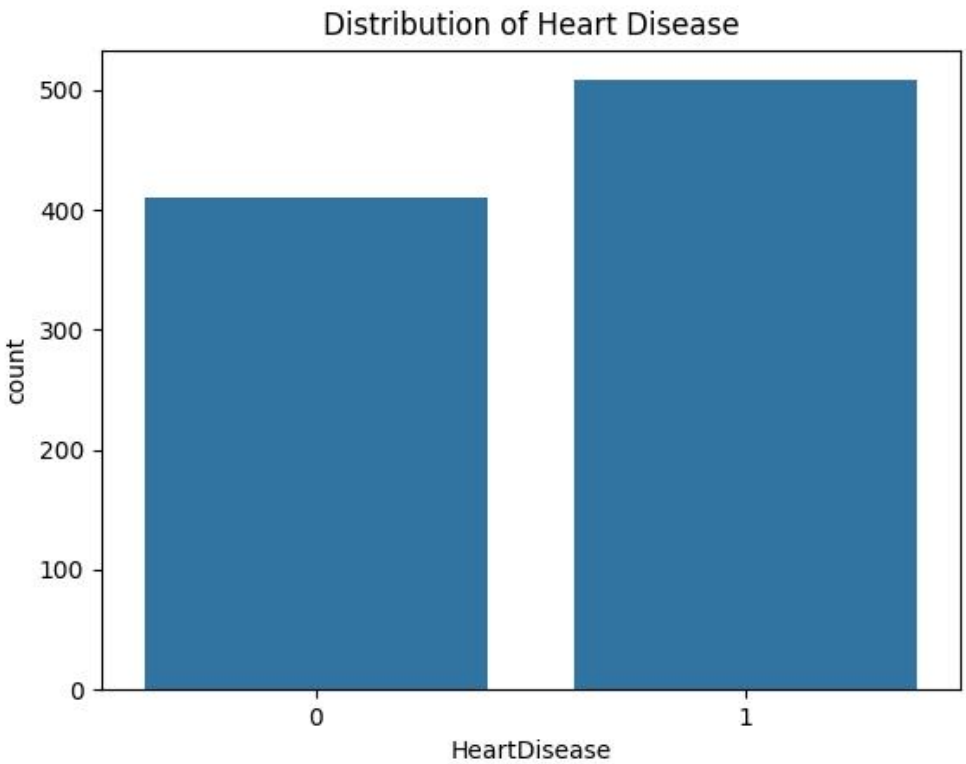


*Figure 1: Distribution of Heart Disease in Dataset*

This figure illustrates the distribution of patients with heart disease (denoted with 1) and without heart disease ( denoted with 0), providing insight into class balance within the dataset.

## 3.1 METHODOLOGY

### Data Preprocessing

Preprocessing the data was done before training the model. Some categorical variables are encoded to numerical values by using the one-hot encoding technique, so that it can be used in a neural network.
The dataset was split into training and testing sets using an 80-20 split.

### Neural Network Architecture

The neural network was implemented using **Keras** with **TensorFlow** as the backend.
A Classification model using Keras with the Sequential API. As I did previously with the input data, I declared an input layer of specific dimension.

The architecture consisted of:
Input layer
Hidden layer (32 neurons)
Hidden layer (16 neurons)
Output layer (1 neuron, sigmoid activation)

The architecture follows a **tapered design** (32 to 16 neurons), which is a strategic choice to facilitate **feature abstraction**. By reducing the dimensionality in the second hidden layer, the network is forced to condense the most relevant clinical patterns from the input data before making a final classification. This structure, combined with the **Adam optimizer**, helps the model navigate the loss landscape efficiently, balancing model capacity with the risk of overfitting on the heart failure dataset.

**Training Procedure**

This model is trained for 30 epochs with the Adam optimiser and binary cross-entropy loss function.
For robustness, the performance of the model was validated on a split during training to assess the model's ability to generalise to unseen data and to identify overfitting.
Three models were trained using different activation functions: Sigmoid, Tanh, and ReLU.

## 4.0 RESULTS

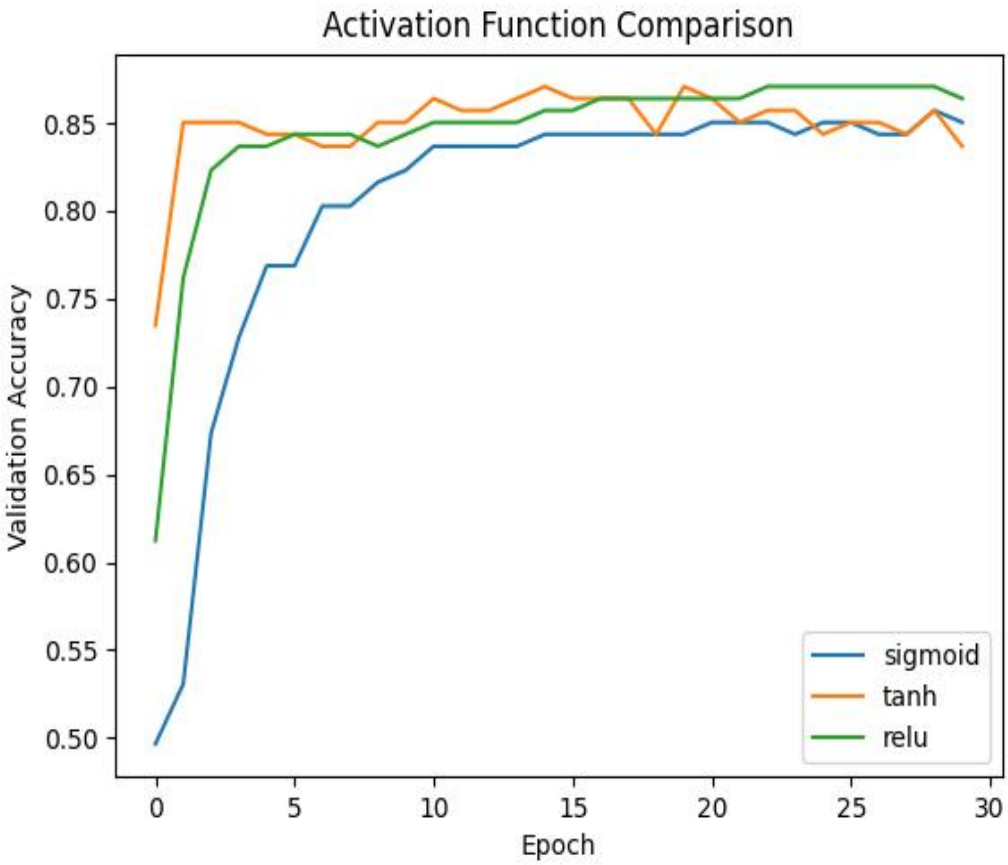


*Figure 2: Validation Accuracy for Activation Functions*

Figure 2 illustrates how validation accuracy evolves during training for each activation function, highlighting differences in convergence speed.

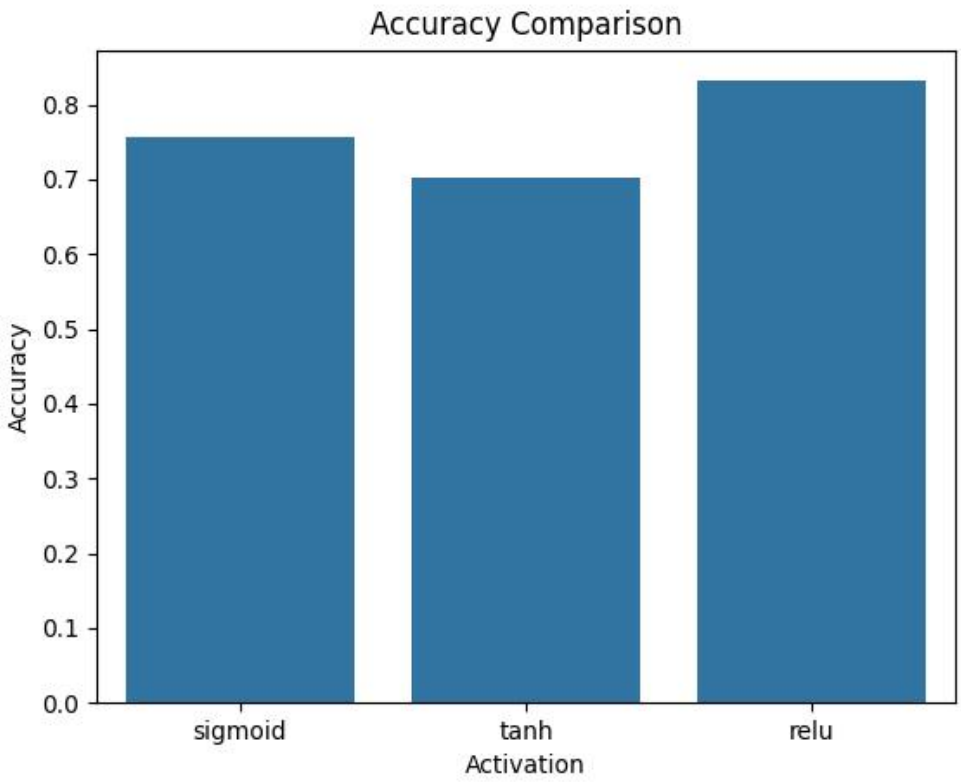


*Figure 3: Accuracy Comparison of Activation Functions*

Figure 3 compares the final classification accuracy across activation functions.

After analyzing these activation functions, the final result is shown in the above figures. It is shown that the accuracy achieved by the ReLU activation function is the highest and the time required to achieve this accuracy is also the shortest. This is due to the non-saturating nature of the gradient of the ReLU function, which means that the weights are updated in the fastest way possible during the backpropagation.

While the previous examples explored the effects of large input values on the derivative for a variety of activation functions, here we show why sigmoid and tanh functions are not a good idea in this context: For large input values, the derivatives are very small, which leads to very slow training. This is yet another factor to consider when choosing an activation function for a neural network.

## 5.0 ETHICAL AI CONSIDERATIONS

Machine learning in healthcare carries a plethora of ethical implications. For one, there is the issue of dataset bias. If a variety of groups are not included in a healthcare training dataset, the resulting machine learning model may not produce clinically accurate, or even fair, results. A study aimed to investigate this problem by generating a simulated 3-D model of the human neck using the SegNet deep learning library in

order to classify neck tumor types with high accuracy. The team experimented with three baseline datasets, which differed in the amount of side information they provided to the model and consequently yielded varying accuracies.
Another concern is clinical responsibility. A machine learning system can be only a tool for a clinician. A prediction by a system may be incorrect and there could be unintended consequences if the system were to be used for any decision making without adequate validation.

As we bring more sophisticated machine learning algorithms into our clinical workflows, issues of equity, transparency and accountability have become some of the most pressing conversations occurring across our industry. Read the full story on FORBES.

## 6.0 CONCLUSIONS

In this tutorial, we have seen how the activation functions used in the neural network can have a profound impact on the accuracy of the model. Here, we were working on a predictive model in the healthcare sector.

I trained a neural network using the Sigmoid, Tanh and ReLU activation functions. The ReLU (Rectified Linear Unit) function gave the best performance as it avoids the vanishing gradient problem and speeds up the learning process.

I would like to know what was important to consider when selecting the activation functions in a deep neural network. According to the findings, the selection of activation functions for a deep neural network is important. Future work could explore deeper architectures and additional optimisation techniques.

**Accessibility Statement**

All figures have labels to show their components and use high contrast colours so that they are easy to read, including for colour blindness. Generally, figures have few components and use short, descriptive titles.

**Github Repository Link:**

https://github.com/brymex-prog/heart-failure-prediction-neural-network

## REFERENCES


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